**Roll No ........................**

# MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **BTech**

Semester: 8 Sem

Name of the Course: Unix Systems Programming

Course code: TCS-826

**Time: 90 Minutes**

**Maximum Marks: 50**

Note:

i. Answer all the questions by choosing any one of the sub questions.
ii. Each question carries 10 marks.

Q1.

(a) Illustrate the working of bash shell when it uses fork() and wait() system calls to run a command on a terminal. Explain with the help of a suitable diagram. [CO1]

OR

(b) Describe Copy on Write (COW) mechanism. How does it help in improving the fork() system call?. [CO1]

Q2.

(a) Differentiate between a mutex and binary semaphore with suitable code example. [CO2]

OR

(b) In Linux systems, what kind of situations are First Come First Serve (FCFS) and Round Robin (RR) scheduling used? Write a small code to show the working of Round Robin scheduling in Linux. *NOTE: RR and FCFS are not used to schedule normal programs.* [CO2]

Q3.

(a) Write C code to implement the following Unix pipeline between two child processes:

cat | more [CO3]

OR

(b) Write C code in which a parent process writes "hello" to a pipe and the child reads the string from the pipe and prints it on the screen. [CO3]

Q4.

(a) Write a code in which the main() thread creates three threads that simultaneously run a function F() while main() waits. Each thread increments a global variable 5000 times so that at the end the global variable contains value of 15000. Note that a thread may need multiple time slices due to scheduling before it could finish incrementing the variable 5000 times. [CO2]

OR

(b) Write a C program using pthreads to create multiple threads, where each thread prints its thread ID while the main() thread waits for all of them to finish before exiting. [CO2]

Q5.

(a) In Message Queue IPC mechanism, what is the use of ftok() call? Write the message structure for Message Queue. [CO2]

OR

(b) Write a program in which a parent process and a child process communicate using Shared Memory IPC mechanism i.e. The child writes a string to shared memory and the parent reads the string and prints it. [CO2]

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